

I: All right. And then I just wanted to start off here with the note regarding you know covid 19 to acknowledge that obviously teaching and learning has been significantly disrupted in the past months. So for any of the questions that we're asking you can answer with reference to your normal teaching practices before this started, or your teaching practices that have been adapted for this situation.

UCSB04: Okay

I: Either are fine. And then just want to start out the interview here with some background. So, if you want to talk about just generally your experience with teaching undergraduate students, especially, especially regarding how your teaching relates to your current and past research, and then which of the courses that you teach, do you actually have students work with data?

UCSB04: Okay. So, there's a variety of courses that I've taught here at UCSB but I've only ever taught here at UCSB. So, I've tried a couple of different types of courses, so I have a larger, upper division undergrad course, which is supposed to be taken sort of between the very large lower division courses and then the smaller upper division courses. And then I've taught a couple of smaller upper division undergrad courses. Couple of grad courses. And then one thing that I particularly had in mind, with regard to this interview and the data use in the classroom is an interdisciplinary honor seminar which is not through my department but it's the [course code redacted].

I: Okay.

UCSB04: And that's very small, it's undergrads but only those who are in the university wide Honors Program whatever that distinction is there with the department level Honors Program. So, in the large upper division undergrad course, it's not so much that I'm using my personal data from my research, should I explain my research?

I: Yeah, you can explain your research, you know, not necessarily what you use in the courses themselves, but just generally your current and past research and how that relates to your teaching or not. As much as you want to get into it.

UCSB04: Yeah, well I figured just for context that could be helpful. So, the data that I collect is on [redacted]. However, when it comes to publishing, I will then sort of, whenever possible, get additional sources of data. For instance, from other researchers other publications, and then combine that with my own data. So, the conclusions I'm trying to reach in my publications are not, I don't want them to just be applicable to my study groups and study species. I want them to be general principles [redacted research topic]. So, the data that I use for this large upper division undergraduate course, it's not the data that I collect from [species redacted] it's more sort of the general patterns that come out from doing the combination of the literature and my own data. And so it's just broader results at that level. And the reason I do that is because [redacted topic] research is driven by a set of hypotheses that are, I think, fundamentally flawed. And this is part of my, what do we call it, I'm trying to give undergrads a different experience than what I had when I was an undergrad, where I had the impression that everything was known, and there was nothing that I could contribute to the field. And so I try to show them the older hypotheses, the gaps in the

hypothesis, and you know how those gaps are illustrated by certain examples from the collated data. And, you know, try to get them to think about what's missing from the original hypothesis, why is it not working, why is, why are there all these species that are not conforming to the predictions? And so that is to get them to think about, you know, this is still a growing and developing field, perhaps they have some contributions to make and kind of trying to get them excited about it.

I: That's really interesting!

UCSB04: So, Yeah, I mean there's this push and pull of getting them interested in the observations, the hypotheses. Thinking constructively about the patterns that are being observed. And at the same time I feel kind of like a hypocrite because doing private field work is extremely challenging and limited. Very few people have the opportunity to actually go out and do that. So on the one hand I'm saying you have something to contribute. On the other hand, it's very much a pay-to-play system and very few people have the resources to get involved in that. So, yeah, So there's that. Then in the smaller upper division undergrad courses, I guess I'm not really using data in those courses so much, but the place where I most intensively integrate data into the courses the [small interdisciplinary honor seminar] and the students go out and collect data on animals in the local campus environment, things like that. And then I teach them how to start looking for patterns in that data, graphing it, making conclusions.

I: Okay, that actually feeds into the next question here, which is more along the lines of how you get data for your courses. So I don't know if you want to expand on that honors course a little more and how they collect data. But the question here is, you know, what's the primary method for how you provide data? Do you have students collect or generate it themselves, search and identify pre-existing data, or do you provide them datasets? And maybe you could expand a little more on that honors course but then also talk about that upper division one as well.

UCSB04: Yeah, okay. So, for the other seminar. I teach them the methods for collecting the data and then we go outside and they have pieces of paper. Basically tables that they fill out while they're observing animals so they are literally collecting their own data. And then I have them enter that data into a spreadsheet, like on Google sheets or Excel, and we go from there so it's very raw, in that sense. Actually there's another small upper division undergrad course that I'd forgotten about which is similar to the honor seminar and that the students collected their own samples that then generate data so this is a lab methods course, where I teach [redacted topic], and assays, and they collect [redacted topic], so certain times a day certain number of days. And then, you know, we start, we've labeled all the urban samples, store them in the lab on campus, and then I teach them how to do [redacted topic]. So again, that stated that they generate and then I teach them how to make sense of that data. First is troubleshooting did they do the assay correctly and then if they did do the assay correctly, what did the results mean for their [redacted topic] so expect to make some inferences from those patterns.

I: Oh, so you teach quite the wide breadth of courses, actually. That's really, that's really interesting too. So, for those two courses that you have students collect data, do you face any challenges with them being able to find or to create those data sets or, it sounds like the process is actually pretty simple for like creating a table.

UCSB04: Oh god no, it sounds simple and for some reason it's just absolutely not. So, I should also say that for the honor seminar, I take a little bit of glee in that it's a bait and switch. So the course is called [discipline topic redacted] and the, you know, the one paragraph description that's on the list of courses for the university is all about [topic/theme redacted]. It says nothing about spreadsheets and statistics. So I use this as a way to get students doing things that they would normally run away from.

I: Okay.

UCSB04: And I have a deep love of data, and my perspective is that it changes how you think. And if they just go through a series of steps, they haven't, you know, made a wrong turn, they haven't understood why it's wrong to do it one way and better to do it a different way. And so I want them to struggle with that process and figure out how to do it correctly. And in the end, they seem to really love it and appreciate that, the frustration was worth it. So, it is my favorite class.

I: It's really interesting to hear because I think often I hear more about people trying to limit those issues that students run into when working with data.

UCSB04: Yeah, Yeah and you know we do have another course. So, again, the honor seminars outside of my department but in my department, specifically in my unit of the department, there's a lower division undergrad course which is meant to prepare them to think about data in this way, and to become used to thinking about how we organize it and set it up so that we can test ideas. And it's very much in the formula as courses go, of a series of steps that the students do where, sure they extensively learn how to calculate a mean from raw data. But for some reason, there seems to be far lower rates of the students really grasping why they're doing what they're doing.

I: Okay.

UCSB04: I actually have not taught that course, so I don't, I personally don't really understand why the honor seminar, I think, seems to be a little bit more effective in that regard. I mean maybe it is a struggle that they go through.

I: And then, so what about the, you have so many courses, it's hard to keep them straight. That upper division course that you do use the monkey data. It sounds like that's data that you provide for them, then

UCSB04: Well so when I say data I guess it's more figures, and in all, it's not the raw data, in that sense, they're a bit more removed from the process in that respect, so they're seeing more the, the final product. Or I'll say for instance, you know, [redacted information about specific assignment]. And, you know, these are some other factors that are involved so it's, yeah I've, I've already collected and processed the data so I know what it says, but I'm trying to lead them to that, that analysis through the figures.

I: So, that course is, it's more about analyzing the data and critically engaging with it. Then, like working with the raw data itself.

UCSB04: Yeah, yeah.

I: Okay. All right. Well, I think I'm gonna move on to the next question, which is more, and I think you've described the process somewhat, but what is the process of students manipulating, analyzing, or interpreting data in your courses including what kind of tools or software they use? And what kind of knowledge, you expect them to come in with if anything?

UCSB04: Well, I have learned to basically have no expectations.

I: Okay.

UCSB04: Yeah, and it's not that I'm being excessively cynical about UCSB students. I remember as an undergrad I did take a statistics course and absolutely none of that made sense to me, even during the course.

I: I had the same experience.

UCSB04: And I did an honors thesis when I was an undergrad, and I had to relearn everything. Basically, there was a grad student who had to use entirely different words and concepts to teach me. And it was frustrating that of course that was supposed to have taught that or helped me learn that. It was so useless but anyway. So the courses here, I have the students using whatever spreadsheet program they have access to and usually that's Google Sheets. And I think it's really important for them not only to collect the data whatever possible, but also to enter the data because that structures a lot of how we use it right like people who've been using data for a decade or three. You know, we know basic things like you can't have a blank row in Excel right or a blank column, you have to have at least one cell with an entry. And it's, it's kind of delightful. What I show them is just little basic things that turned out to make things so much easier like pivot tables in Excel. Because I try to tell them over and over and over again that even if they're not going to be an academic and go off to grad school and, you know, work with data for the rest of their lives, it's really useful to understand programs like Google sheets or Excel because that's a skill that you can take to many many, many other jobs. So it's a translatable skill. And, yeah, so in the process of entering their data they always make a bunch of mistakes and then we go back and we fix them. And then we start sort of creating reports through pivot tables and inevitably there's a couple of students who don't understand what I'm saying, when I'm trying to, when I put up on the screen, you know how we go from a raw table to a pivot table and sort of create these summaries, and they end up doing it by hand. And then when they finally do start using the pivot table, that sense that they have of, oh my god this is so much easier, you know, why aren't we learning this in elementary school.

I: Yeah.

UCSB04: Yeah. So, they do all have that. And I think that that part is almost as important as learning how to chunk the data, calculate averages and either standard deviations or standard errors of the mean and then graphing it. And then I teach them, you know, whether it's a bar graph or a line plot with, you know, confidence intervals. I teach them how to sort of do an eyeball scan of that figure to interpret. Is there likely to be a significant pattern here, and I'm using the old sense of statistical significance.

I: Okay. And do you find, other than the ways in which you've already highlighted, do you find that there are other ways that students are challenged in working with the data?

UCSB04: They have the syndrome where they kind of can't control their hands. I don't know if you've ever seen this but they, when they're not quite sure what to do or they're not, they may be a little distracted, their hands are sort of flicking around on the mouse pad. And so, the spreadsheet is flying around, and it's like this negative feedback loop where once that spreadsheet is flying around, and they can't see like if they forget to freeze that top row so they don't, they don't even know what columns they're looking at. They then start to panic, that they don't know what to do and they're behind, and they're fucking everything up and etc. So, and this is why I haven't even tried to teach this course in the online version like you know I'm waiting until we can do in person classes again because that's when I say, take your hands away from the keyboard.

I: Yeah.

UCSB04: And, you know, start over. Tell me the steps and why you're doing each step. But there's this physical feedback with the hands and flipping around that I can't even wrap my head around. I mean, I don't know where that comes from.

I: I suppose and then they click something by accident and aren't seeing the same thing you are.

UCSB04: And you know it's one thing to sort of click around between programs but even within like if they're using Excel, they seem to not initially grasp the difference between a click, click and drag, a double click. All kinds of crazy things happening because they've just dragged a column over somewhere else and replaced the column, you know, and had no idea what they were doing.

I: Okay, I see. Interesting. Yeah like the scrolling or the accidental manipulation, that, yeah, that you don't know how to undo or they panic and don't realize they can undo it.

UCSB04: Yeah, exactly.

I: I see, and then moving on here. How do you see the ways that, I think you've touched on this already when talking about your research and your motivation for your courses but, how do you see the ways in which you teach data relating to the learning goals in your field?

UCSB04: It's a subset of the goals, and I think this is why in some of my classes I don't have them working with data at all, because those are a different set of goals, and the classes where I have integrated data, especially the extensive forms in the lab class and the honor seminar, that is the hands on learning experience. So the classes where I'm not using data. You know, these are classes where it's more of a big picture overview because, so for instance like the primary [redacted class topic], large upper division undergrad course, it's typically about 100 students, and especially this year I have more students who are not [major redacted] majors they're coming from [major redacted] and [major redacted] and [major redacted] and all these different departments.

I: And do you think that's just because the courses that they might have taken in their own departments are not available remotely?

UCSB04: So, enrollment was flagging within [discipline redacted], for this particular course, and so I made a flyer, and I distributed that all over campus so that's why I have more. Departments represented in the courses, this quarter but yeah so the basic prerequisite for the courses that the students have taken some kind of intro to evolution and should, in theory, understand the mechanics of that, you know what is natural selection, how does it work, how do we know that natural selection is occurring in a population, things like that. And the thing about, of course like [redacted class topic] is that you know there's depending on how you count, maybe 400 or 500 [redacted class topic] species in the world. And the reason that we study [redacted class topic] to begin with is sort of twofold. One is that they're our closest relatives and so we learned about, at least in this course [redacted class topic]. And so it's a great sort of testing ground for all kinds of theories. And so for that large upper division undergrad course, the data that they see is already sort of processed and set up in a way to get them thinking because, you know I am trying to keep a big picture. Many, many [redacted class topic] species, many different patterns. With the courses where they are actually collecting data, learning how to enter it and summarize it and interpret it, that's a very narrow window. So we're not talking about hundreds of [redacted class topic] species we're talking about one species at a time. And how do we start from the bottom up, start from the observations and how do we structure our observations in a way that we can actually test the research question. So, yeah, top down versus bottom up perspectives.

I: And do you generally find that your courses that are focusing on the raw, the raw data skills are smaller than the bigger picture ones?

UCSB04: Absolutely. The way that I teach them is very hands on for them and hands on for me because I have to do so much troubleshooting with them, you know, dealing with the frustration dealing with the inadvertent effects of twitchy hands. Yeah, there's you know scrolling frantically through pages and not thinking about what they're looking at. And I mean it yeah I wish that all my classes could be small, but I also wish that every student had the experience of being in something like that honors seminar.

I: And, and I know you touched on this with the Excel skills but do you also see that you are teaching them specific data skills that will directly prepare them for future careers either academic or otherwise?

UCSB04: So the skills that I'm hoping that they learn that they can then apply to future jobs or studies is more about how do they figure out how to deal with a problem. Right, so maybe they're never again going to need to calculate and the standard error or the mean. But just sort of understanding that, A, they need a way to represent the data, and, you know, B, how do they go about figuring out the best way to do that without a statistics textbook at hand necessarily. So it's the problem solving I want them to learn.

I: Yeah. And that part is so important, or just like thinking about things and in context of data. I know I've talked to grad students who you know are in PhD programs that don't have a grasp of that even. It's such an important thing. All right. And have you observed any policies or cultural changes at your

institute, here at UCSB, or in your department specifically that influenced the ways that you teach with data? I don't know how, how many years you've been teaching these courses and if you've adjusted them based on policy changes or cultural changes.

UCSB04: I don't think that I've really seen policy changes. It's more for me, it's less of a temporal change and more of a difference between organizational units so my department versus the honors program right like there's a different approach there.

I: So the honors program on campus here has their own approach for teaching their courses or for teaching with data specifically?

UCSB04: Just in the sense of course size. So I would really have to justify teaching such a small class within [discipline redacted], and it's not that we have a ton of majors, but we don't have enough classes for the majors that we have. Anyway, the other sort of motivation for me personally for teaching the honor seminar, of course, is that there's like a little research stipend that comes with it for the faculty, and I can't use it for salary but I can use that to fund my research, and my research is constant and expensive, relatively speaking. It's a nice additional source of income.

I: Yeah. You know I'm not really aware of the Honors Program at all here yet. Even though I've been here a year I'm still learning a lot about UCSB. And then, do you find just generally in your fields that instructors or researchers in general face ethical challenges when teaching with data? And that's similar to the last question but yeah.

UCSB04: I don't know of any instances personally but I wouldn't be surprised if there were some just because. So, you know, again I'm an animal behavior researcher. I don't really study humans. But my colleagues do study humans, and you know they're applying for IRBs all the time and dealing with that. So I'm aware that there are those types of issues. I will say, in terms of ethical issues, whether it's data, or even papers that students are producing. You know, and each new quarter, students often want examples of what's expected of them, and they want to see, for instance, essays from the previous year. What's a good essay or, you know, what does the finished product look like when you're dealing with the data, and I don't want to show them something from a previous class because I haven't gotten permission from those students in the previous quarter to show it and future quarters. So there's that aspect of it.

I: And is that a concern with you for sharing data from previous quarters too?

UCSB04: Oh yeah, no I don't share it at all. So, yeah.

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I: So, and I imagine that specifically with like, these challenges aren't as important to your teaching because as you said you're not. Well, other than the students, you're not working with human data. That's with animal data.

UCSB04: Yeah. Now the lab course. I've only taught that once. It was, it's something that we're supposed to now be teaching every year but we're trying to hire an instructor for that but the course

is quite a lot of work. And in the course of the students collecting their own samples and labeling them and assaying them and collecting your, you know interpreting the data. I did sort of have at the back of my mind. Oh, what happens if we see that so what happens if we see that might be embarrassing for a student. And thankfully in that first quarter there, there weren't any such examples but I guess there's a possibility for that in the future so...

I: That thought popped into my mind too, so are all of the samples available to that, like, does the whole class see the results of everyone's sample or is that person's sample private to them?

UCSB04: Well, that would be one option but I for various reasons I have them working in small groups within a small group they do see each other's data and typically in class what was happening is that, you know, a student would run into a problem and therefore this class time that was dedicated to ironing out these issues and so often they would share their screen in class and say you know this is what I saw how did I do this wrong and so you know we'd walk through the troubleshooting together so that, those are instances where they were showing their, their data to the class and it was fine in that quarter because nothing unusual was happening but again in future that might potentially be problematic.

I: Yeah, yeah that's such an interesting, like it's such an interesting set of data to be working with, I think I was talking with someone previously and they were talking about the importance of data that you have a personal connection to. But I could see a lot of questions coming up from that as well, or issues.

UCSB04: Yeah. Yeah. And you know, on the one hand, even if there's nothing unusual in the data, you know students might be hesitant, or think twice about showing the results of the class. On the other hand, they are so deeply invested in the process because it's their own samples and it's a level of interest that I haven't seen from students in other contexts. So I'm reluctant to give up that level of student involvement.

I: There's a sense of ownership there since it is their own data.

UCSB04: Yeah, yeah. And even when they don't understand something they want to understand it. Because theirs, you know.

I: Yeah, that's so interesting I could see the same thing with DNA, as well. I mean, I think it's just something that everyone's intrinsically interested in. But anyway, I can move on.

UCSB04: But I thought I'd say one last thing there, so because we're starting off with urine samples. You know, they're starting with a pool of students who are not grossed out by this stuff right. They are happy to talk for hours about how like, Oh, I thought it would be easy but then it turns out like peeing into a tiny little cup is not so simple, because I peed all over my hand so how am I going to deal with that for the future. Or like, You know my pee was super dark and then you know this other person said well my pee was kind of a weird color should I be concerned and, you know, there's just all kinds of aspects to it that I had not anticipated because again this is the first time I had taught that course but again really had them invested in understanding what was going on with their own bodies.

I: Yeah, I wouldn't have expected that, but I suppose, maybe the shared experience too, but they're all doing this makes people more open. Yeah, about their experiences.

UCSB04: That's true.

I: Alright. One of the final portions here is about training and support in the courses. So do you have anyone other than you providing instruction or support, TA's, other instructors, librarians, or drop in sessions anywhere on campus?

UCSB04: So, when I'm doing the bulk of it, because the data intensive courses are small, I don't have a TA. However, during that honors seminar where the students observe animals on campus and then collect data and analyze it. I do every quarter have a one class meeting where we go to the library. In the past, Gosh, I don't even know how many times I've taught the course but there've been one or three times maybe where they've met with a librarian. But, I guess it was the last time I taught the course we actually had sort of a, what do you call that one, the library agrees to do a training session like a workshop?

I: We usually call them one-shots, instruction sessions.

UCSB04: Yeah. So ,we did a more formal one the last time.

I: And it was out with a subject librarian or someone in the Collaboratory.?

UCSB04: Yeah, I think it was a subject like librarian but, and so the last time I taught that course was, Oh gosh. So, this year's 2020-2021. It would have either been early 2019 or late 2018 because last year I was on sabbatical. And that's why my memories are a little bit fuzzy here but so we had this sort of formal workshop with the subject librarian, and it was useful but the students were a little bit tuned out. And I think, to me it was interesting because it was more like a traditional class, and they had, most of the time with the class we were reading outside even if we are analyzing data. And so I think they kind of resented having to be put into a classroom.

I: I see, yeah, I know that's something that we struggle with is, you know, setting the right tone or making sure we're talking about things that actually, you know, relate directly to the things that students are doing in their courses and it's hard if you haven't been in the course other than that one day.

UCSB04: Now every time I teach the course we do at least go to the library for one of the class meetings, maybe two, and then my reason for that as each year I'm horrified, I ask the students, How many of you have ever gone to the library and of course everybody raises their hand because they've studied in the library, say okay so how many of you use the library and of course they'll raise their hands again because to them that's using the library for studying, right, I say, Okay, how many of you have ever checked a book out. And, on average, it's not more than one student per quarter and usually it's maybe every other quarter.

I: No more than one student per quarter? Like one student out of your cohort, your course? Wow.

UCSB04: And then I say okay so you've checked a book out. Did you check the book out from this library, and they say no, they check the book out from their public library back home.

I: interesting.

UCSB04: I say okay so for all of you guys regardless of whether you've checked a book out. How do you figure out what book to check out, and they're like well we don't look for books, because, you know, unless it's online we don't care about. So, the class meeting where we're in the library and I started off with a series of questions. I say okay well I'm going to force you guys to, you know, use, I can't even remember what it's called the library website where you look up materials online. Yeah, you have to find the call number and then you have to find the book on the shelf and once you find the book on the shelf, you have to stay in that area for like half an hour and look at the other books around that book, because maybe that book is not actually going to be useful to you but you don't know until you pull it off the shelf and start looking at it. And then at the end of class you're actually going to check out the book and take it home. And, yeah, so that's the first time they've ever done that and I feel like a grumpy old lady, but I don't care.

I: That's so interesting with, like, it feels like just how quickly things have moved online, too. I mean, even between my undergrad program and my graduate program, I used the library all the time. And then I used to almost, I mean, even as a library student I barely used it because there were so many materials online.

UCSB04: Yeah. Yeah. And part of the reason that I have them go through this whole process is not just that, you know I want them to have the experience of checking out a book and holding a book in their hands and looking for information in a physical book. But time and time again, they realize that the material that they have access to online is partial, they're not getting the full picture and the stuff that they're looking up in these books about the Natural History of the animal that they have chosen to study for the course. The species. And in the course of looking up the, you know, finding the book and determining whether it's useful and then you know looking at the other books around, they find out how much information they are missing because it's not online.

I: Yeah, that's really important. Especially just the free resources online and there's so many things behind paywalls then too and then yeah the books, the physical books in the library.

UCSB04: Yeah, well, and this is a field where people were observing animals for decades, right before stuff went online and a lot of that stuff is in books, it's not even in journal articles. And those books have not been digitized so that's, I want them to get a sense of what's known about the Natural History of the species before they even start collecting data. And I want them. I want the book, and what's known to inform how they collect the data,

I: I see. Yeah, that's really interesting. I'm really shocked I mean I'm, I'm surprised but I'm also not surprised by the one per quarter number.

UCSB04: Yeah, yeah, no I mean I, I've never had more, I mean that's a small class right but this is the class I've taught most often, actually.

I: So, Do you find that students are learning to work with data outside their formal coursework either through things like, you know, like I'm thinking like data or not data camp, what is like coding, like different coding resources or data resources online, or from other internships or things like that before they come into the course or do they typically only have course experience with data.

UCSB04: So, no, no. I've never met a student who had attended something that was not basically a formal class right for course credit, and of the students in the class, you know, over the many quarters, only one or two had ever taken something like a statistics course with data. And, you know, they had the same experience that I did where they were like yeah I know I was supposed to learn stuff and I did stuff but I don't. I didn't absorb it. And the thing about the honors seminar is that, you know, these are students not coming from just one major one department, they're all over the university. And part of the reason that I don't say in the course description that they will learn how to enter and graph and analyze and interpret data is that that's the sort of thing that a lot of students from social sciences and from the humanities will run away from.

I: Oh, yeah, I can attest to that. I didn't take a single statistics course until I was in grad school. I actively avoided it.

UCSB04: Yeah, exactly. And so it's easier for them to sort of get involved with it and then for me to say, guess what guys you're actually doing statistics and they go, No. And, you know, I have them doing this because I want them to lose the fear, you know, rather than thinking, Oh, this is something that I can't do. It's just part of course and not a big deal I guess yeah I mean it is a big deal but it's not.

I: And so do you encourage them to do any kind of other extracurricular learning outside of the course or after your course.

UCSB04: So in the honor seminar, not really because you know that it's not actually a full credit course right it's only two units instead of four. So even there, I don't really give them much in the way of homework or readings. Because it's supposed to be less of a demand on their time. It's just enough that they make sure to show up every week because they're interested and you know they're there. To give you an example. The more convenient species to study on campus are the turtles in that little pond between the [student center] and and the tower, and almost none of the students come into the course thinking that turtles are interesting and by the end of the quarter, they've given every turtle a name, and they are highly invested in these as individuals. And, you know, they, they talk about it like a soap opera right like so and so did this did the other one and then you know this is how they responded and you know he's such a bully and blah blah blah blah right so so they do get invested in it. But, yeah, so if a student has an interest for instance in doing field research in ecology or animal behavior or thinking about grad school, I give them lots of information about that. But in terms of pursuing other opportunities nobody's really asked like how do I do more data analysis or how do I do more statistics.

I: Okay. And then, as far as your own training goes, have you received training in data, other than your graduate degree like have you done other workshops or training, since that formal training?

UCSB04: Yeah, so at UCSB just the last, I guess couple years so, so this year I did one of those data carpentry workshops for Python right, done a workshop for R, and I'm frustrated with myself because it didn't stick, but the reason for that is that I've been using STATA since mid grad school, and it makes sense to me and obviously Excel. It's not that I want to keep using Excel. But I have found over and over and over....Sorry.

I: Yeah. No, I was gonna say it takes a lot to learn a new language, I mean it's a different language, in the end.

UCSB04: Yeah, no I mean that's that's absolutely part of it I'm, I'm really short on time right so it's not like I have the luxury of doing a 12 hour dive into any one new software, or language. But maybe this is just a quirk of this, the grad students that I have experienced with and exposure to. But the students who adopt R or Python or something like that early on and do their utmost to avoid Excel, they make crazy mistakes. Right, so. Granted, I'm not super familiar with those software programs so basically they'll join data sets or manipulate them and then do analysis on them and come up with these amazing results in their analyses and when we backtrack and sort of check the flow of the data handling. I always discover some really, really serious fundamental problems. And my interpretation of this is that when they're using R, especially it's kind of like a black box and so they learn the code for doing a step and then they don't check it to make sure that the output makes sense.

I: Or they don't understand what's going on behind the code.

UCSB04: Exactly. And I think that when they learn how to do the analysis using Excel where they're actually looking at the numbers, it's a lot easier to catch those mistakes. So, I'm torn because I feel like a Neanderthal for not knowing how to use R properly at all, really. And it's what everybody uses everybody in the social sciences and at least the younger generations. And I want to be able to reduce, you know, the, the nice looking figures and to use the same language that everybody's using so that, you know, I can post my code with every manuscript and people can look at it and say yeah that's that's robust, I know that she did this right but at the same time, I am not letting go of the old methods because time and time again, I've seen how important it is to be physically keyed into the data at every step and to check it and think about what's happening. And then, I'm trying to teach that to my current grad students but it's a struggle.

I: Yeah, no that makes sense. Especially considering, you know, when you use R and Python that's usually at the step after doing the data collection. So after you've used Excel or whatever other collection method and it's hard I know with our workshops too. We try to give a grasp of everything, you know the collection portion and. And then, like how a spreadsheet should look, but it's hard to cover everything and of course the code part is like the flashy part right yeah. So, that's really interesting.

UCSB04: One of my goals during my sabbatical was to spend some time with R and that did not happen. That's my own fault.

I: That's okay and it's been, I can't imagine you have any spare time right now. I feel like I haven't had any and I can't imagine what it's like for people teaching courses right now. Let's see. Last thing

I want to ask you, do you use any, like, data sets, or syllabi or other instructional resources that you received from other people or do you make them all yourself and do share those with others?

UCSB04: So, let's see. For my big upper division undergrad course I inherited the lectures from somebody else. And there are problems with them, and I keep meaning to make it my own and fix all those problems but that lack of time, lack of full interest. The smaller upper division undergrad courses and [discipline redacted] and the honors seminar, those are things that I have designed myself and I've shared the materials, both for the course that I inherited, and for at least one, maybe two of the other courses with people, and other universities like new faculty who have been hired and stuff like that.

I: Okay. And then I have one more question for you, and we'll wrap up here. Sorry if I'm keeping you over time. What is, considering the trends in your field, what types of training or assistance do you think would be most beneficial to instructors like yourself with teaching data?

UCSB04: So the thing that I'm frustrated with every year, and that it would be lovely if there were a better workaround for this. When students are doing the deep dive into data so they're collecting the data, entering, analyzing it, graphing etc. Most of the students don't have Excel on their personal computers and the only way that they could, that I know of, that they can have access to Excel is by going into one of the computer labs. And, you know, I don't want to lose the outdoor aspect of the class right because that's, it's the class that they look forward to because they know that they're going to be sitting in the sun and, you know, talking with members of their small group etc. So that means that a lot of the students are forced to use Google sheets which is super limited in its functions. And it's clunky. Yeah.

I: This might be an ignorant question since I am new here but don't, the students aren't able to get Microsoft Office with Excel for free, or is it just that they have computers that can't support it?

UCSB04: I don't think that they get it for free, I think they have to pay.

I: Oh, okay. I'll have to take a look at that. I've never had that experience at any other institution I've been at so

UCSB04: Yeah, welcome to UCSB. So yeah, there's probably a workaround for that. But I think that's the one thing that sort of immediately comes to mind. I do keep the course kind of basic and lo fi, just because I don't want to get, I don't want to spend too much time on fancy tech and have the students be a step removed from the really hands on process of, for instance, entering their data and stuff.

I: Yeah, and obviously that brings in other issues and I mean there are always installation errors galore with carpentry workshops, yeah so that's a whole nother set of issues to have to deal with, and troubleshoot. Okay. Well, is there anything else that you want to share what you thought of since the beginning here.

UCSB04: So, you know if there's ever a push to make that sort of sort of like a universal curriculum aspect of having students working with data from the beginning of their time in college, I'd be 1,000% behind that.

I: Yeah, I know once in a while we throw around the idea of an INT course for data management, but I think that would be even, even a larger push than that.

UCSB04: And I know I said this at the beginning of the interview but it's worth repeating, working with data changes your brain. And the students appreciate it once they've been sort of pushed into that experience or, you know, had a little bit of hand holding. And, you know, if you want to make sort of an argument that it's justified because it's useful to the students right like there's so many different careers where any little bit of handling data will improve their skill set improve their understanding of the patterns and things like that that they're seeing with that data that they're working with, big data is all the rage right, those are the jobs, in data science.

I: Thank you.